



Enhanced role of graphitic-N on nitrogen-doped porous carbon ball for direct dehydrogenation of ethylbenzene

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ABSTRACT

Nitrogen doping is an efficient strategy to improve the catalytic activity of carbocatalysts in direct dehydrogenation reactions. The interpretation of enhanced activity is now mainly attributed to the strengthened basicity. On the other side of doping, electronic properties that differed in various nitrogen doping configurations, however, were barely considered. Herein, to elucidate the role of graphitic N, pyridinic N and pyrrolic N, N-doped porous carbon balls with tunable contents of these nitrogen species were used as moulded carbocatalysts for steam-free direct dehydrogenation of ethylbenzene to styrene. Along with C=O groups that were commonly certified as active sites on carbocatalysts, graphitic N contributes to the styrene yield. The former serves as active site that directly activates and dissociates C–H bond, while the latter one provides additional electrons into the delocalized π -system and leads to improved chemical reactivity of C=O. The loss of C=O active sites is not only attributed to the carbon deposition as illustrated by the elevated area ratio of D₃ to G band in Raman spectra, but also its failed regeneration from intermediate hydroxyl groups (C–OH), which was proved by the tiny thermal decomposed products of CO and H₂O as detected by the on-line mass spectrometry and gas chromatography.

1. Introduction

Styrene (ST) is one of the most fundamental chemicals in plastic industry, and metal-free carbocatalysis of steam-free direct dehydrogenation (DDH) of ethylbenzene (EB) has attracted great attentions [1–4]. Unsaturated carbonyl groups (C=O) on the carbon material are widely considered as the active center in DDH reaction. The substantial electron density at the oxygen atom makes it a Lewis base site to activate the electron-poor saturated C–H bond. Then carbonyl groups (> C=O) abstract the hydrogen atoms and transform to hydroxyl groups (> C–OH). After the regeneration of C=O from C–OH at high temperatures, active sites finish one catalytic cycle [5,6].

Recently, nitrogen-doping is proved as an effective strategy to improve the DDH catalytic activity of carbon materials [1–3,7]. N-doped carbon nanotube and the designed N-doped carbon/carbon nanotube composite foam (N–C/CNT) all exhibited over 1.5 times ST production rate in comparison to the N-free ones [1,7]. The enhancement is traditionally attributed to the strengthened basicity. Generally, nitrogen doping configurations are diverse in the nitrogen-doped carbon

material including graphitic N, pyridinic N and pyrrolic N. Li et al. studied the individual basicity of these nitrogen species by employing DFT calculations, which suggested that pyridinic N has a strong Lewis basicity, followed by pyrrolic N, while graphitic N delivers almost no enhancements to the basicity [8]. Nitrogen atom has one more valence electron than carbon. Thus, besides changes of basicity, the incorporation of nitrogen also can tailor the electronic properties of carbon materials with redistributed electron density [1,9–11]. Evidently, different nitrogen doping configurations possess distinct chemical environments and electronic effects. Pyrrolic N and pyridinic N located at the cavities or edges donate two and one p-electrons to constitute the aromatic six π -electron system, respectively; particularly, pyridinic N possesses one lone electron pair in the sp^2 hybrid orbital [12,13]. Graphitic N refers to the nitrogen substituting a carbon in the graphitic structure and contributes one more p-electron to the π -system. Their distinct contribution, which has been verified in oxygen reduction/evolution reactions [12,14,15], can also be expected in other reactions. However, the question that which nitrogen doping configuration is related to the active site of nitrogen-doped carbon materials in

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